Practice Sheet: this, call/apply/bind, Prototypes & Classes

30 Questions (Easy → Moderate → Hard)

🟢 EASY LEVEL (Q1–Q10)

These questions are focused on understanding concepts.

Question 1 — Method and this

```
const user= {
name:"Ritik",
greet() {
console.log(this.name);
}
};
user.greet();
```

Expected Output

Ritik

What is this question asking?

Identify what this refers to when a method is called using an object.

Concepts Tested

- this
- Method Calls

Question 2 — Default Binding

```
function show() {
console.log(this);
}
show();
```

Expected Output

Depends on:

- Browser
- Node.js
- Strict Mode

Concepts Tested

- Default Binding
- this

Question 3 — Object Method

```
const car= {
brand:"BMW",
showBrand() {
console.log(this.brand);
}
};
```

Predict output.

Concepts Tested

- this
- Object Methods

Question 4 — call()

```
function greet() {
```

```
console.log(this.name);
}
const user= {
name:"Aman"
};
```

Use call() to print:

Aman

Concepts Tested

- call()

Question 5 — apply()

```
function introduce(city) {
console.log(`${this.name} from${city}`);
}
const person= {
name:"Ritik"
};
```

Use apply().

Expected Output

Ritik from Bhopal

Concepts Tested

- apply()

Question 6 — bind()

```
function greet() {
console.log(this.name);
}
const user= {
```

name:"Priya"

};

Create a permanently bound function.

Concepts Tested

- bind()

Question 7 — Arrow Function this

const user= {

name:"Ritik",

greet: () ⇒ {

console.log(this.name);

}

};

user.greet();

Predict output.

Concepts Tested

- Arrow Functions
- Lexical this

Question 8 — Event Handler Theory

What is the difference between:

button.addEventListener("click",function(){})

and

button.addEventListener("click", () ⇒ {})

regarding this?

Concepts Tested

- Event Handlers

- this

Question 9 — Constructor Function

Create:

function Person(name) {

this.name=name;

}

Create an object:

new Person("Ritik")

Expected Output

{

name:"Ritik"

}

Concepts Tested

- Constructor Functions

Question 10 — Prototype Lookup

const arr= [1,2,3];

Where does:

arr.push()

come from?

Concepts Tested

- Prototype Chain

🟡 MODERATE LEVEL (Q11–Q20)

Question 11 — Lost this

```
const user= {
name:"Ritik",
greet() {
console.log(this.name);
}
};
const fn =
user.greet;
fn();
```

Fix it.

Expected Output

Ritik

Concepts Tested

- bind()

Question 12 — Borrow Method Using call()

```
const user1= {
name:"Ritik"
};
constuser2= {
name:"Aman"
};
```

Create one method and use it for both objects.

Concepts Tested

- call()

Question 13 — Apply with Math.max

Find maximum number using:

Math.max()

and:

apply()

Array:

[10,20,50,5]

Expected Output

50

Concepts Tested

- apply()

Question 14 — Create Prototype Inheritance

const animal= {

eats:true

};

Create:

dog

using:

Object.create()

Concepts Tested

- Object.create()

Question 15 — Shared Prototype Method

Create:

Person.prototype.greet

and make multiple objects use it.

Concepts Tested

- Prototype
- Memory Optimization

Question 16 — Check Prototype Relationship

function Person() {}

const p = newPerson();

Verify:

p.__proto__===Person.prototype

Expected Output

true

Concepts Tested

- **proto**
- prototype

Question 17 — Student Class

Create:

class Student

with:

name

marks

Concepts Tested

- Classes
- Constructor

Question 18 — Car Class

Create:

class Car

with:

brand

start()

Output:

BMW started

Concepts Tested

- Classes
- Methods

Question 19 — Getter

Create:

fullName

using a getter.

Example

p.fullName

should return:

Ritik Rajput

Concepts Tested

- Getters

Question 20 — Setter

Create:

fullName

setter.

Input:

p.fullName="Aman Gupta"

Expected Behavior

Update:

firstName

lastName

Concepts Tested

- Setters

🔴 HARD LEVEL (Q21–Q30)

Question 21 — Employee Inheritance

Create:

class Employee

and

class Developer

using:

extends

Concepts Tested

- Inheritance

Question 22 — Animal Hierarchy

Create:

Animal

↓

Dog

↓

Labrador

Scenario

Multi-level inheritance.

Concepts Tested

- Prototype Chain
- Classes

Question 23 — Static Method

Create:

MathHelper.add()

Example

MathHelper.add(5,10)

Output:

15

Concepts Tested

- Static Methods

Question 24 — Static Property

Create:

MathHelper.PI

Output:

3.14159

Concepts Tested

- Static Properties

Question 25 — Bank Account

Create:

class BankAccount

with:

#balance

Methods:

deposit()

withdraw()

Expected Output

700

after:

deposit(1000)

withdraw(300)

Concepts Tested

- Private Fields

Question 26 — Shopping Cart Class

Create:

class ShoppingCart

Methods:

addItem()

removeItem()

getTotal()

Scenario

E-commerce Application.

Concepts Tested

- Classes
- Arrays
- Objects

Question 27 — Library Management System

Create:

Book

Library

classes.

Methods:

addBook()

borrowBook()

returnBook()

Concepts Tested

- Classes
- OOP Design

Question 28 — Prototype vs Class Investigation

Create:

class Person {}

Verify:

typeof Person

Expected Output

function

What is this question asking?

Prove that classes are syntactic sugar.

Concepts Tested

- Classes

- Prototypes

Question 29 — Custom Array Method

Add:

Array.prototype.last

Example:

[1,2,3].last()

Output:

3

Concepts Tested

- Prototype Extension

Question 30 — Complete School Management System (Interview-Level)

Create:

class Student

class Teacher

class School

Requirements:

Student

name

marks

Teacher

name

subject

School

Methods:

addStudent()

addTeacher()

getTopper()

getAverageMarks()

Example Output

Topper:Ritik

AverageMarks:82

What is this question asking?

This is a mini project combining:

- this
- Classes
- Constructors
- Methods
- Arrays
- Objects
- Inheritance (optional)
- Real-world Design